

Career Roadmap: Data Scientist

Generated by AI Career Coach | 3 Month Plan

Week 1: Python for Data Science Fundamentals

Leverage existing dev skills to master pandas, numpy, and EDA concepts.

- Install Jupyter, Pandas, Numpy, Matplotlib, Seaborn (1h) - Beginner
- Complete Pandas 101 exercises (filtering, grouping, merging) (4h) - Beginner
- Analyze a Kaggle dataset and create 3 basic plots (5h) - Intermediate

Week 2: Advanced EDA & Data Cleaning

Learn to handle missing data, outliers, and feature engineering.

- Write a script to clean a dirty dataset (handle NaNs, drop duplicates) (3h) - Beginner
- Apply One-Hot Encoding and StandardScaler to categorical/numeric features (3h) - Intermediate
- Build an EDA notebook that generates a data quality report (4h) - Intermediate

Week 3: Practical Statistics & Probability

Build the mathematical intuition required for data science.

- Calculate mean, median, variance, and standard deviation manually then with Numpy (2h) - Beginner
- Simulate a normal distribution and calculate z-scores (3h) - Intermediate
- Perform a basic T-test on two sample groups using Scipy (5h) - Intermediate

Week 4: A/B Testing & Experimentation

Apply statistics to real-world business problems via A/B testing.

- Define metrics for a mock website conversion test (2h) - Beginner
- Calculate required sample size for a 5 percent Minimum Detectable Effect (3h) - Intermediate
- Analyze a mock A/B test dataset and determine statistical significance (5h) - Advanced

Week 5: Introduction to Machine Learning

Transition from rules-based coding to predictive modeling using Scikit-Learn.

- Split a dataset into training and testing sets (1h) - Beginner
- Build a Linear Regression model to predict house prices (4h) - Intermediate
- Build a Logistic Regression model for binary classification (5h) - Intermediate

Week 6: Model Evaluation & Tuning

Understand how to measure model performance and optimize hyperparameters.

- Implement K-Fold Cross Validation on your Logistic Regression model (3h) - Intermediate
- Calculate Precision, Recall, and plot an ROC curve (4h) - Intermediate
- Use GridSearchCV to find the best hyperparameters for a model (3h) - Advanced

Week 7: Tree-Based Models

Learn powerful non-linear algorithms widely used in industry.

- Train a Decision Tree and visualize the splits (3h) - Intermediate
- Train a Random Forest classifier to improve accuracy (4h) - Intermediate
- Extract and plot feature importances from the Random Forest (3h) - Intermediate

Week 8: Gradient Boosting & Advanced ML

Master state-of-the-art tabular data algorithms.

- Train an XGBoost model on a complex dataset (4h) - Advanced
- Apply SMOTE to handle an imbalanced classification problem (3h) - Advanced
- Compare XGBoost, Random Forest, and Logistic Regression performance (3h) - Intermediate

Week 9: Unsupervised Learning

Find patterns in data without explicit labels.

- Implement K-Means clustering for customer segmentation (4h) - Intermediate
- Use the Elbow method to find the optimal number of clusters (2h) - Intermediate
- Apply PCA to reduce a high-dimensional dataset to 2D for plotting (4h) - Advanced

Week 10: Natural Language Processing (NLP) Basics

Process and analyze text data.

- Clean text data (remove stopwords, punctuation, lowercase) (2h) - Beginner
- Convert text to features using TF-IDF Vectorizer (3h) - Intermediate
- Train a Naive Bayes classifier for sentiment analysis (5h) - Intermediate

Week 11: Deep Learning Foundations

Introduction to Neural Networks using PyTorch.

- Complete PyTorch 60-minute blitz (Tensors and Autograd) (3h) - Intermediate
- Build a basic Multi-Layer Perceptron (MLP) for classification (4h) - Advanced
- Train the MLP and plot the training loss curve (3h) - Advanced

Week 12: Model Deployment & Portfolio Building

Leverage your dev background to deploy ML models.

- Save a trained model using joblib or pickle (1h) - Beginner
- Wrap the model in a FastAPI endpoint (4h) - Intermediate
- Build a simple Streamlit frontend for the API (5h) - Advanced